

Math 230B Lecture Notes

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1 Week 5

Example ($SL(n, \mathbb{F})$). Suppose $\mathbb{F} = \mathbb{R}$ (or $\mathbb{C}$). We claim that

$$SL(n, \mathbb{F}) = \{A \in GL(n, \mathbb{F}) : \det A = 1\}$$

is a Lie subgroup of $GL(n, \mathbb{F})$.

Indeed, we have that

- $SL(n, \mathbb{F})$ is closed under multiplication, indeed, if $A, B \in SL(n, \mathbb{F})$, then

$$\det(AB) = (\det A)(\det B) = 1 \cdot 1 = 1.$$

So, AB is invertible and $\det(AB) = 1$. This implies that

$$AB \in SL(n, \mathbb{F}).$$

- $SL(n, \mathbb{F})$ contains the inverse of each of its elements. Indeed, suppose $A \in SL(n, \mathbb{F})$, then A is invertible and

$$\det(A^{-1}) = \frac{1}{\det A} = \frac{1}{1} = 1.$$

So, $A^{-1} \in SL(n, \mathbb{F})$.

From the above, we can conclude that $SL(n, \mathbb{F})$ is a subgroup of $GL(n, \mathbb{F})$.

2 Week 6

Example (Special Orthogonal Group). Note that

$$\begin{aligned} SO(n) &= \{A \in M(n, \mathbb{R}) : A^T A = I \text{ and } \det A > 0\} \\ &= \{A \in M(n, \mathbb{R}) : A^T A = I \text{ and } \det A = 1\} \end{aligned}$$

We will prove that $SO(n)$ is a subgroup of $O(n)$.

- Note that $SO(n)$ is closed under multiplication. Indeed, suppose A and B are in $SO(n)$. Our goal is to show that $AB \in SO(n)$. Then we have

$$A, B \in SO(n) \implies A, B \in O(n) \implies AB \in O(n).$$

Since $A, B \in \text{SO}(n)$, it follows that $\det A = 1$ and $\det B = 1$. Hence,

$$\det AB = (\det A)(\det B) = 1 \cdot 1 = 1.$$

- Also observe that $\text{SO}(n)$ contains the inverse of each of its elements. Indeed, if $A \in \text{SO}(n)$, then

$$A \in \text{SO}(n) \implies A \in \text{O}(n) \implies A^{-1} \in \text{O}(n). \quad (1)$$

Furthermore,

$$A \in \text{SO}(n) \implies \det A = 1 \implies \det A^{-1} = \frac{1}{1} = 1 \implies A^{-1} \in \text{SO}(n) \quad (2)$$

Definition (Bilinear Form). A function $B : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ is said to be a **bilinear form** on $\mathbb{R}^n$ if it satisfies the following condition:

$$(1) \quad \forall x, y, z \in \mathbb{R}^n \quad B(x + y, z) = B(x, z) + B(y, z)$$

$$(2) \quad \forall c \in \mathbb{R} \text{ and } \forall x, y \in \mathbb{R}^n \quad B(cx, y) = cB(x, y)$$

$$(3) \quad \forall x, y, z \in \mathbb{R}^n \quad B(x, y + z) = B(x, y) + B(x, z)$$

$$(4) \quad \forall c \in \mathbb{R} \quad \forall x, y \in \mathbb{R}^n \quad B(x, cy) = cB(x, y)$$

Moreover,

- The bilinear form B is said to be **symmetric** if

$$\forall x, y \in \mathbb{R}^n \quad B(x, y) = B(y, x).$$

- The bilinear form B is said to be **skew-symmetric** if

$$\forall x, y \in \mathbb{R}^n \quad B(x, y) = -B(y, x).$$

- The bilinear form B is said to be **non-degenerate** if the following statement is true

$$B(x, y) = 0 \quad \forall y \in \mathbb{R}^n \implies x = 0.$$

- A symmetric bilinear form B is said to be **positive definite** if

$$B(x, x) > 0 \quad \forall x \neq 0.$$

Example (Standard Inner Product on $\mathbb{R}^n$). The standard inner product in $\mathbb{R}^n$

$$\langle x, y \rangle = \sum_{i=1}^n x_i y_i$$

is a symmetric positive definite bilinear form on $\mathbb{R}^n$.

For example, let's verify condition (1) for bilinearity:

$$\begin{aligned} \langle x + y, z \rangle &= \sum_{i=1}^n [(x_i + y_i)z_i] \\ &= \sum_{i=1}^n [x_i z_i + y_i z_i] \\ &= \sum_{i=1}^n x_i z_i + \sum_{i=1}^n y_i z_i \\ &= \langle x, z \rangle + \langle y, z \rangle. \end{aligned}$$

I leave you to check the other conditions yourself.

In general, a symmetric positive definite bilinear form on a real vector space V , by definition, is called an **inner product** on V .

Example (Standard Symplectic Form on $\mathbb{R}^n$). Define $w : \mathbb{R}^{2n} \times \mathbb{R}^{2n} \rightarrow \mathbb{R}$ as follows:

$$\forall x, y \in \mathbb{R}^{2n} \quad w(x, y) = \sum_{j=1}^n x_j y_{n+j} - x_{n+j} y_j.$$

For example,

$$\begin{aligned} n = 1 &\implies w(x, y) = x_1 y_2 + x_2 y_1 \\ n = 2 &\implies w(x, y) = (x_1 y_3 - x_3 y_1) + (x_2 y_4 - x_4 y_2) \\ &\vdots \end{aligned}$$

If we let $\Omega = \begin{pmatrix} O_{n \times n} & I_{n \times n} \\ -I_{n \times n} & O_{n \times n} \end{pmatrix}$, then we can write the standard symplectic form in the following way:

$$w(x, y) = \langle x, \Omega y \rangle$$

where $\langle, \rangle$ is the standard inner product in $\mathbb{R}^{2n}$.

Proposition (Preserving the standard symplectic form on $\mathbb{R}^{2n}$). Suppose $A \in M(2n, \mathbb{R})$. We say that A preserves the standard symplectic form on $\mathbb{R}^{2n}$ if and only if $A^T \Omega A = \Omega$.

Proof. We say that $A \in M(2n, \mathbb{R})$ preserves the standard symplectic form on $\mathbb{R}^{2n}$ if and only if

$$\begin{aligned} \forall x, y \in \mathbb{R}^{2n} \quad w(Ax, Ay) = w(x, y) &\iff \langle Ax, \Omega(Ay) \rangle = \langle x, \Omega y \rangle \\ &\iff \langle x, A^T \Omega Ay \rangle = \langle x, \Omega y \rangle \\ &\iff \forall y \in \mathbb{R}^{2n} \quad A^T \Omega Ay = \Omega y \\ &\iff A^T \Omega A = \Omega. \end{aligned}$$

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Proposition. If $A^T \Omega A = \Omega$, then A is invertible.

Proof. Suppose $A^T \Omega A = \Omega$, then we have

$$\begin{aligned} \det(A^T \Omega A) &= \det(\Omega) \\ \implies \det(A^T) \det(\Omega) \det(A) &= \det(\Omega) \\ \implies (\det(A))^2 \det(\Omega) &= \det(\Omega) \\ \implies (\det(A))^2 &= 1 \\ \implies \det(A) &= \pm 1 \\ \implies A &\text{ is invertible.} \end{aligned}$$

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Proposition. $A^T \Omega A = \Omega$ if and only if $-\Omega A^T \Omega = A^{-1}$ where A is invertible.

Proof. We have

$$\begin{aligned}
 -\Omega A^T \Omega = A^{-1} & \iff -A^T \Omega = \Omega^{-1} A^{-1} \\
 & \text{left multiply by } \Omega^{-1} \\
 & \iff -A^T \Omega A = \Omega^{-1} \\
 & \text{right multiply by } A \\
 & \iff -A^T \Omega A = \Omega \\
 & \iff A^T \Omega A = \Omega.
 \end{aligned}$$

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Example (Real Symplectic Group). $\text{Sp}(2n, \mathbb{R}) = \{A \in M(2n, \mathbb{R}) : A^T \Omega A = \Omega\}$ is a subgroup of $\text{GL}(2n, \mathbb{R})$.

Proof. We just proved that every matrix in $\text{SP}(2n, \mathbb{R})$ is invertible. So,

$$\text{SP}(2n, \mathbb{R}) \subseteq \text{GL}(2n, \mathbb{R}).$$

- We prove that $\text{SP}(2n, \mathbb{R})$ is closed under multiplication. Suppose $A, B \in \text{SP}(2n, \mathbb{R})$. We have

$$\begin{aligned}
 (AB)^T \Omega (AB) &= B^T (A^T \Omega A) B \\
 &= B^T \Omega B \\
 &= \Omega.
 \end{aligned}$$

Hence, we have $AB \in \text{SP}(2n, \mathbb{R})$.

- $\text{SP}(2n, \mathbb{R})$ contains the inverse of each of its elements. Suppose $A \in \text{SP}(2n, \mathbb{R})$. We have

$$\begin{aligned}
 A^T \Omega A = A &\implies (A^T)^{-1} \Omega A = (A^T)^{-1} \Omega \\
 &\implies \Omega = (A^T)^{-1} \Omega A^{-1} \\
 &\implies \Omega = (A^{-1})^T \Omega A^{-1} \\
 &\implies A^{-1} \in \text{SP}(2n, \mathbb{R})
 \end{aligned}$$

From the above, we can see that $\text{SP}(2n, \mathbb{R})$ is a subgroup of $\text{GL}(2n, \mathbb{R})$. Now, we will prove that $\text{SP}(2n, \mathbb{R})$ is a Lie subgroup of $\text{GL}(2n, \mathbb{R})$. Define the map

$$G : \text{GL}(2n, \mathbb{R}) \subseteq \mathbb{R}^{4n^2} \rightarrow M(2n, \mathbb{R}) \cong \mathbb{R}^{4n^2} \text{ by } A \mapsto A^T \Omega A - \Omega.$$

Clearly, $\text{SP}(2n, \mathbb{R}) = F^{-1}(\{0\})$. It can be shown that

$$\forall A \in \text{GL}(2n, \mathbb{R}) \quad DF(A) \text{ has rank } \frac{4n^2 - 2n}{2} = 2n^2 - n.$$

Hence, $\text{SP}(2n, \mathbb{R})$ is an embedded submanifold of $\text{GL}(2n, \mathbb{R})$ with dimension $4n^2 - (2n^2 - n) = 2n^2 + n$. Thus, $\text{SP}(2n, \mathbb{R})$ is a Lie subgroup of $\text{GL}(2n, \mathbb{R})$. ■

Definition (Homomorphisms, Isomorphisms, Automorphisms). Let G and H be two groups.

- A function $\varphi : G \rightarrow H$ is said to be a **homomorphism** if it preserves group multiplication

$$\forall g_1, g_2 \in G \quad \varphi(g_1 g_2) = \varphi(g_1) \varphi(g_2).$$

- A bijective homomorphism is called an **isomorphism**. If there is an Isomorphism between G and H , we say G and H are isomorphic, and we write $G \cong H$.
- An **endomorphism** of G is a homomorphism from G to itself.

- An **automorphism** of G is an isomorphism from G to itself.

Definition (Lie Group Homomorphism). Let G and H be two Lie groups.

- A function $f : G \rightarrow H$ is said to be a **Lie group homomorphism** if
 - f is a group homomorphism
 - f is smooth
- A function $f : G \rightarrow H$ is said to be a **Lie group isomorphism** if
 - f is a group isomorphism
 - f is smooth
 - f^{-1} is smooth

If f is smooth and its inverse is smooth, then f is a diffeomorphism.

Example (The Determinant is a Lie Group Homomorphism). The determinant $\det : \text{GL}(n, \mathbb{R}) \rightarrow \mathbb{R}^*$ is a Lie Group Homomorphism:

- $\forall A, B \in \text{GL}(n, \mathbb{R})$, we have $\det(AB) = \det(A)\det(B)$. Hence, $\det$ is a group homomorphism.
- $\det A$ is a polynomial in terms of the entries of A and so differentiable. Hence, the $\det$ is smooth.

Similarly, the $\det \text{GL}(n, \mathbb{C}) \rightarrow \mathbb{C}^*$ is a Lie Group Homomorphism.

Example (Left Translation/right translation). Let G be a Lie Group and let $g \in G$. The **left translation by g** is the map

$$L_g : G \rightarrow G \quad h \mapsto gh.$$

The **right translation by g** is the map

$$R_g : G \rightarrow G \quad h \mapsto hg.$$

Notice that L_g is a composition of smooth maps and so it is smooth:

$$G \rightarrow G \times G \rightarrow G \quad h \mapsto (g, h) \mapsto gh.$$

Starting from left to right, we see that the first mapping is smooth because

- Fact 1: $F = (F_1, F_2)$ is smooth if and only if F_1 and F_2 are smooth.
- Fact 2: $h \mapsto g$ is a constant function implies that it is smooth.
- Fact 3: $h \mapsto h$ is the identity which is also smooth.

The second mapping $G \times G \rightarrow G$ is smooth because multiplication in G is smooth since G is a Lie group.

In fact, $L_g : G \rightarrow G$ is a diffeomorphism because the inverse L_g is $L_{g^{-1}}$:

$$\forall h \quad L_{g^{-1}} \circ L_g(h) = g^{-1}(gh) = h.$$

Unfortunately, L_g in general is NOT a group homomorphism; that is, in general

$$L_g(h_1 h_2) \neq L_g(h_1) L_g(h_2).$$

Similarly, the right translation map also satisfies the properties above but is NOT a group homomorphism.

Example (Conjugation by $g \in G$). Let G be a Lie group and let $g \in G$. The conjugation by g is the map:

$$C_g : G \rightarrow G \quad h \mapsto ghg^{-1}.$$

Note that C_g is a diffeomorphism because it is a composition of diffeomorphisms:

$$C_g = L_g \circ R_{g^{-1}}.$$

In fact, $C_g^{-1} = C_{g^{-1}}$. Also, note that

$$\forall h_1, h_2 \in G \quad C_g(h_1 h_2) = g(h_1 h_2)g^{-1} = (gh_1 g^{-1})(gh_2 g^{-1}) = C_g(h_1)C_g(h_2).$$

Hence, $C_g : G \rightarrow G$ is a Lie group isomorphism (i.e a Lie group automorphism).